

GOURAV KAR

Java Backend Developer | Spring Boot | Python & Machine Learning

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Portfolio: gouravkar.vercel.app

PROFESSIONAL SUMMARY

Computer Science Engineering graduate (2026) with hands-on backend development experience in Java, Spring Boot, and Python, building REST APIs and full-stack applications. Comfortable working across the stack with FastAPI, Next.js/React, and SQL Server, and applying Machine Learning and LLM integration (Scikit-learn, Google Gemini, OpenRouter) to build AI-assisted applications. Gained Agile/Scrum exposure through a backend development internship at Acceleratron, with a working foundation in REST API design and system design.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript

Backend & APIs: Spring Boot, FastAPI, REST APIs, Pydantic, Swagger/OpenAPI, Microservices, JPA, Hibernate

Frontend: React, Next.js, Streamlit, Tailwind CSS

Machine Learning: Scikit-learn, Pandas, NumPy, Feature Engineering, Model Deployment

AI Integration: Google Gemini API, OpenRouter / LLM APIs, Generative AI, Agentic AI workflows

Databases: SQL Server, SQLite

Cloud & Deployment: Docker, Render, GitHub Pages

Testing: Pytest, Vitest, Playwright

DevOps & Tools: Git, GitHub, SourceTree, Maven, npm, Jira, Confluence

Methodologies: Agile, Scrum, System Design, Architecture Planning, Estimation

Other: Gmail SMTP Automation, Windows OS Scripting

WORK EXPERIENCE

Backend Developer Intern — Acceleratron

February 2024 – June 2026

- Developed and maintained backend components using Java, contributing REST API endpoints and Spring Boot services for internal business systems
- Contributed to a garment industry ERP initiative, supporting inventory, sales, and transaction-tracking functionality
- Participated in Agile sprints and Scrum ceremonies, contributing to sprint planning and task estimation
- Received a team management award in recognition of contributions during the Acceleratron engagement

PROJECTS

FinNexus — AI Trading Workstation | [GitHub](#)

2026

Python, FastAPI, Next.js, TypeScript, Tailwind CSS, SQLite, Docker, OpenRouter API

- Built a full-stack trading workstation with a FastAPI backend serving REST APIs and Server-Sent Events market streaming, alongside a statically-exported Next.js/TypeScript frontend, all served from a single containerized service
- Implemented a real-time market data pipeline combining live price ingestion with a Geometric Brownian Motion price simulator, streaming updates to the frontend via SSE
- Developed a portfolio engine supporting trade execution, position and cost-basis tracking, and a treemap-based portfolio heatmap visualization
- Integrated an LLM-powered AI copilot via the OpenRouter API that parses natural language trading commands into structured actions and auto-executes them
- Containerized the application with Docker and covered backend and end-to-end flows with Pytest and Playwright test suites

PARTHI-ERP — Garment Industry ERP System | [GitHub](#)

June 2025 – August 2025

Java 17, Hibernate / JPA, SQL Server (JDBC), Maven, Swing

- Designed and developed a Java desktop ERP application for the garment industry, using Hibernate/JPA over a SQL Server database for inventory, sales, and transaction data
- Built a Swing-based desktop interface to support stakeholder and client workflows, including stock tracking and transaction record-keeping
- Structured the codebase as a multi-module Maven project (core and agency modules), separating business logic from data access

- Used Lombok and SLF4J/Logback to reduce boilerplate and support application-level logging

AI-Based Maternal Health & Preeclampsia Risk Prediction System | [GitHub](#)

2026

Python, Scikit-learn, Pandas, NumPy, Streamlit, Plotly, Google Gemini API, Render, GitHub Pages

- Built a full-stack ML web application with a dual-stage prediction pipeline: a maternal health risk model that escalates high-risk cases to a dedicated preeclampsia risk model
- Engineered the end-to-end ML workflow — data cleaning, feature engineering, model training, and evaluation — using Scikit-learn, Pandas, and NumPy on clinical datasets from Zenodo and Hugging Face
- Built an interactive Streamlit dashboard with Plotly visualizations, gauge-style risk indicators, and an emergency symptom flagging feature
- Integrated the Google Gemini API to power a conversational AI assistant offering maternal health guidance based on prediction results
- Implemented automated PDF report generation and deployed the application on Render with a companion landing page on GitHub Pages

FlowKanban — Full-Stack Kanban Board with AI Assistance | [GitHub](#)

2026

Python, FastAPI, Next.js, React, Docker, OpenRouter API

- Built a full-stack Kanban board application with a FastAPI backend and a Next.js/React frontend communicating via REST APIs
- Implemented drag-and-drop task management with full undo/redo history tracking on the frontend
- Integrated an AI board assistant powered by the OpenRouter API to support automated board actions
- Containerized the application with Docker and validated it with backend Pytest tests, frontend Vitest tests, and Playwright end-to-end tests

PreLegal — Legal Agreement Drafting Platform | [GitHub](#)

2026

Next.js, React, TypeScript, Tailwind CSS, Vitest

- Built a Next.js/React/TypeScript web application for drafting, customizing, and exporting standard legal agreements based on the open-source Common Paper template library
- Implemented an interactive form builder with multi-tab party/term configuration, live document preview, and variable highlighting to audit custom terms against boilerplate
- Built PDF export (html2canvas + jsPDF), print-optimized styling, and Markdown clipboard export for sharing drafted agreements
- Covered the application with a Vitest and React Testing Library unit/integration test suite

Additional Projects: A2Z Fast Food — terminal-based restaurant ordering service built around a Python REST API ([GitHub](#)). Login Alert — Python script that emails real-time Windows login/unlock alerts via Gmail SMTP ([GitHub](#)).

EDUCATION

B.Tech in Computer Science Engineering

Bengal Institute of Technology

August 2023 – July 2026

Diploma in Computer Science Engineering

Raghunathpur Government Polytechnic

June 2021 – August 2023

10th Grade — WBBSE

Kamalpur Netaji High School

February 2020

CERTIFICATIONS & TRAINING

- Java Training
- Spring Boot Training
- Artificial Intelligence by IBM
- Python Programming with Application on Robotics
- AWS Academy Cloud Foundation
- SAP S/4HANA Development Certification